INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

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Scope, role and assumptions

I am writing as the BC/DR Coordinator for Northbridge Skills Institute. At 08:00 WAT on Day 0 (a Monday), flooding and a power failure made the main site and its local systems unavailable, and Facilities expects the closure to last 72 hours. No cyberattack is confirmed. Even so, the backup design has weaknesses, so I treat possible compromise as something to rule out, not something to assume away.

Where the brief is silent I made assumptions and I label them: **A1** the B02 chain holds hourly incrementals from 00:00 to 07:00, because the catalogue only gives the endpoints; **A2** examination master copies and accessibility-adjustment records exist outside the flooded site, for example in Microsoft 365; **A3** the alternate facility has a room that can seat the 40 learners who need a workstation; **A4** maturity is rated on a five-level scale whose names I map to the slide 13 model in Task 6; **A5** all times are WAT and "elapsed" means time since 08:00 on Day 0. Anything about backups I could not verify is labelled **U** (unverified).

Task 1: Business Impact Analysis

My first job at 08:00 was to work out what the outage actually threatens, and in what order. F1 to F4 each need the recovery network, AUTH01, DB01 and their own application, while F5 needs only the network, AUTH01 and FS01. Table 1 records ownership, people, premises, suppliers, information and dependencies. Table 2 shows how impact grows over time and the minimum service I will accept while systems are down (NIST, 2010).

Table 1: BIA profile of F1 to F5

Function	Owner role	Users affected	People and premises	Suppliers and information	Dependencies
F1 Teaching and assessment (LMS01, DB01)	Head of Teaching and Assessment	1,800 learners; 120 sitting the 11:00 exam; teaching staff, invigilators	Invigilators, assessment lead, 2 technicians; alternate-site room with 25 workstations	Alternate-site provider, Microsoft 365; exam masters, registers, accessibility adjustments (A2)	Network, AUTH01, DB01, LMS01; staff, secure access, communication
F2 Fees and finance (FIN01, DB01)	Finance Lead	Fee-paying learners; finance staff	Finance officer and a separate approver	External payment provider (available), bank; provider settlement reports, fee ledger, wallet balances	Network, AUTH01, DB01, FIN01; staff, secure access, communication
F3 Admissions and registration (REG01, DB01)	Registrar	Applicants, new and continuing learners; registry staff	Registry officers and an approver	Application forms, supporting documents, registration change requests	Network, AUTH01, DB01, REG01; staff, secure access, communication
F4 Staff payroll (HR01, DB01)	HR and Payroll Manager	All 90 staff	HR officer, finance approver	Bank submission portal; payroll master data, approved change list, last bank file	Network, AUTH01, DB01, HR01; staff, secure access, communication
F5 Records and certificates (FS01)	Records Manager	Learners, alumni, employers requesting verification	Records officers	Certificate templates and stock; request	Network, AUTH01, FS01; staff, secure access, communication

				register; restricted record files	

Table 2: Impact over time, minimum service and priority

Function	Impact 0 to 8 h	8 to 24 h	24 to 72 h	Minimum acceptable service and controls	Priority
F1	Exam at 11:00 at risk; NGN 120,000 trigger; integrity and access concerns	Teaching lost; assessment backlog	Coursework deadlines slip; complaints; awarding body concern	Every learner gets a valid same-day sitting: 80 own-device via controlled Microsoft 365 delivery, 25 at stations, 15 in a supervised later sitting on an equivalent version. Head of Assessment approves each version; attendance sheets; submissions saved to a restricted location with a file hash; marks held until LMS01 is validated.	1
F2	Fee confirmation due 16:00; local receipts and wallet cannot update	Unreconciled receipts; learner disputes	Wrongly applied access blocks; payment complaints	The provider portal is the payment record. Manual fee log in a restricted file; preparer plus separate approver; provisional confirmation only; no wallet changes or refunds until reconciliation.	2
F3	Applications cannot be entered; admissions close 18:00	Lost applicants and enrolment	Registration queue; reputation loss	Online application form with time-stamps and the same fields; Registrar approves each registration change in a log; no deletions; unique reference per entry.	3
F4	Low immediate impact; 12:00 Day 1 bank deadline approaches	Risk of wrong or duplicate instructions	Staff pay errors; manual correction	Use restored HR01 once validated. If not validated by 08:00 Day 1, submit the last approved payroll plus a signed change list; HR	4

				manager and finance approver both sign; check against the previous bank file for duplicates.	
F5	None urgent	48-hour commitment under pressure	Commitment breached; verification delays	Requests logged in a restricted register; urgent certificates only with Registrar approval; no records on personal devices; access limits unchanged.	5

Impact calculations

Each cost is the hourly rate multiplied by hours of disruption, over the first 24 hours with no recovery or continuity measures. For F1, 60,000 × 8 = NGN 480,000 and 60,000 × 24 = NGN 1,440,000. The same method gives the other lines in Table 3. The five rates add to NGN 135,000 per hour, so the base impact is 135,000 × 8 = NGN 1,080,000 at eight hours and 135,000 × 24 = NGN 3,240,000 at 24 hours. With no measures in place there is no acceptable assessment service at 11:00, so I add the NGN 120,000 rescheduling cost once. The totals are NGN 1,200,000 at 8 hours and NGN 3,360,000 at 24 hours.

Table 3: Estimated disruption cost with no recovery or continuity measures (NGN)

Function	**Function**	**8 hours (rate × 8)**	**24 hours (rate × 24)**
F1 Teaching and assessment	60,000	480,000	1,440,000
F2 Fees and finance	40,000	320,000	960,000
F3 Admissions and registration	20,000	160,000	480,000
F4 Staff payroll	10,000	80,000	240,000
F5 Records and certificates	5,000	40,000	120,000
Subtotal	135,000	1,080,000	3,240,000
One-off examination rescheduling (trigger met, counted once)	n/a	120,000	120,000
Combined total		**1,200,000**	**3,360,000**

These are exercise estimates, not expected losses, and they have two main limits. First, a flat hourly rate spreads cost evenly, but real harm arrives in steps at deadlines: a missed 16:00 fee confirmation or an 18:00 admissions closure costs far more than one hour's average. Treating the lines as non-overlapping also hides knock-on effects, such as a rescheduled exam delaying fee decisions. Second, the figures count only incremental disruption costs. They leave out recovery spending such as alternate-site hire and overtime, any cost after hour 24, and the likelihood of the event.

Money is also only part of the picture. Learner welfare (stress, unfair outcomes, travel in flood conditions), reputation with learners and employers, and data integrity (wrong payroll, lost payments, altered files) have no reliable hourly price, yet they can outlast the outage.

Priorities, deadlines and dependencies

The 11:00 examination shapes the whole morning. It starts three hours after the disruption, and the baseline recovery in Task 5 restores LMS01 only at 11:45. The exam therefore cannot wait for technical recovery, so I launch the assessment continuity arrangement immediately and treat restoration as the route back to normal, not the route to the exam. Later deadlines set the business order: F1 (11:00 exam), F2 (fees by 16:00), F3 (admissions close 18:00), F4 (bank deadline 12:00 on Day 1) and F5 (48-hour commitment).

Business priority is not technical restoration order. Shared infrastructure fixes the technical sequence: network, AUTH01, security and backup checks, DB01, then the four applications. FS01 supports the lowest-priority function, yet it needs only AUTH01 and the checks, so it can be restored while DB01 runs and the second technician would otherwise sit idle. F5 therefore returns before F1 for technical reasons. Among the applications I follow business priority, starting with LMS01. Four dependency risks could break the plan:

- **AUTH01 is a single local directory**, separate from Microsoft 365, and its snapshot is eight hours old. If it is corrupt or compromised, all five functions are blocked.
- **DB01 is shared by four functions.** One 90-minute restore and one backup chain carry F1 to F4, and the B02 chain depends on every incremental being intact (A1).
- **Production and backup administration share one identity**, and the offsite provider's access independence is undocumented (U). One stolen or misused credential could damage both copies.
- **Capacity and connectivity are thin.** There are two technicians, 25 workstations against a need for 40, and mobile internet is the only route to Microsoft 365.

Task 2: Recovery objectives and backup design

I use three terms consistently (ICDFA, Week 21; NIST, 2010). The **Recovery Time Objective (RTO)** is the target time to restore a function to its agreed minimum service. The **Recovery Point Objective (RPO)** is the maximum acceptable age of recoverable data, measured back from the moment of disruption. The **Maximum Tolerable Downtime (MTD)** is the longest disruption the business can bear before damage becomes unacceptable. Targets are commitments I must justify and test, not hopes, so each RTO leaves a buffer below its MTD for validation, business acceptance and things going wrong.

Table 4: Recovery objective matrix (times from 08:00 on Day 0)

Function	RTO	MTD	**Buffer (MTD minus RTO)**	RPO	**Basis for the numbers**
F1	4 h (12:00); exam continuity service by 10:45	8 h (16:00)	4 h	30 min	Assessment integrity and learner records are sensitive; MTD stops short of a second teaching day lost.
F2	5 h (13:00)	8 h (16:00)	3 h	30 min	Fee confirmation due 16:00; payment records are the hardest to rebuild.
F3	6 h (14:00)	10 h (18:00)	4 h	1 h	Admissions close 18:00; registrations can be re-captured from forms.
F4	8 h (16:00)	24 h (08:00 Day 1)	16 h	4 h	Bank deadline is 12:00 Day 1; four hours kept for verification and submission.

F5	12 h (20:00)	36 h (20:00 Day 1)	24 h	8 h	48-hour commitment less a 12 h safety margin.
Supporting: AUTH01	Within F1 to F5 RTOs	n/a	n/a	4 h	Account and permission changes must be recent for access controls to hold.

I will declare a function recovered only when it passes the service level in its minimum-service definition, not when the restore job finishes. That means: restore complete; integrity and reconciliation checks passed; authorised users can sign in with correct permissions; and the function owner has signed an acceptance record. I allow 15 to 30 minutes for acceptance and count it inside the buffer.

Checkpoint ages and data-loss windows

Age is the disruption time (08:00 Day 0) minus the checkpoint time, so each age below is a data-loss window. Table 5 compares them with the RPOs above.

Table 5: Age of latest recoverable checkpoints at 08:00 and gap to RPO

Items	**Checkpoint**	**Calculation**	**Age**	**Functions**	**RPO versus age**
DB01 (B01 + B02 chain)	07:00 Day 0	08:00 minus 07:00	1 h	F1 to F4	F1 30 min: gap 30 min. F2 30 min: gap 30 min. F3 1 h: met, no margin. F4 4 h: met, 3 h margin.
DB01 full only (B01)	23:00 Day −1	08:00 minus 23:00	9 h	F1 to F4	Fallback if the chain breaks: misses every RPO.
AUTH01 (B03)	00:00 Day 0	08:00 minus 00:00	8 h	F1 to F5 access	RPO 4 h: gap 4 h.
FS01 (B04)	23:00 Day −1	08:00 minus 23:00	9 h	F5	RPO 8 h: gap 1 h.
Disconnected full	17:00 Day −3	08:00 minus 17:00 Day −3 (72 h − 9 h)	63 h	All (last resort)	Far outside every RPO; clean-copy reference only (U).

The windows are measured at the moment of disruption. Work done on local systems during the outage is lost on top of them unless temporary records capture it. Some data can be reconciled: payments from the provider's transaction reports, which are an independent source; applications and registration requests from emails, forms and phone logs; exam attendance from invigilator sheets and Microsoft 365; and payroll changes from signed forms. I cannot assume recoverable anything held only on local systems: receipts and wallet changes between 07:00 and 08:00, marks entered locally, account or password changes after 00:00, and file changes after 23:00 on FS01. I also cannot assume the backups themselves are clean or complete (U).

Backup design

The current gaps come from infrequent checkpoints and from an untested, non-independent chain. My design in Table 6 closes the RPO gaps and applies the 3-2-1 principle with an extra independent copy (NCSC, n.d.). Copy 1 is production data. Copy 2 is an on-site repository on separate disks, kept above flood level or in a second building. Copy 3 is an encrypted offsite copy on different media, and a fourth immutable or offline copy sits in a separate failure domain. Together that gives three copies, two media types, one offsite and one immutable.

Table 6: Proposed backup schedule and controls

Item	**Schedule**	**Retention**	**Placement and controls**
DB01	Full nightly 23:00; differential every 4 h (03:00, 07:00, 11:00 …); incremental every 15 min	14 daily, 8 weekly, 12 monthly	On-site repository, encrypted offsite copy, weekly full to immutable or offline storage (30 to 90 day lock).
AUTH01	Snapshot every 4 h and after any directory change	30 days	Same three placements; configuration export stored with the immutable copy.
FS01	Full nightly 23:00; incremental every 4 h	14 daily, 8 weekly, 12 monthly	Same placements; access lists backed up with the files.
All sets	Verification after every job	Logs kept 13 months	Encryption (AES-256 or equivalent) in transit and at rest; keys held separately under two-person control; separate backup administrator identity with MFA and a sealed break-glass account; alert on any backup administrator logon; restore tests below.

Restore chain for DB01 today: B01 (full, 23:00 Day −1), then every incremental in order, 00:00 to 07:00 (eight items under A1). If any link is missing or fails verification I stop at the last intact link and accept a larger data-loss window. **Under the new design** at an 08:00 disruption the chain is the 23:00 full, the 07:00 differential, and the incrementals at 07:15, 07:30 and 07:45, giving a latest recoverable point of 07:45, a 15-minute window.

An offsite copy or live replica alone does not establish recoverability. An offsite copy proves only that a file exists: it may be incomplete, encrypted with a key nobody can reach, or deletable by the same compromised administrator. A live replica copies corruption, deletion and malicious changes within seconds. Recoverability needs three checks. For **integrity**, I verify checksums of every chain link, reconcile the catalogue against the provider's inventory, and scan for malware. For **access independence**, I confirm that no production credential can delete or alter backups, retrieve a copy with the separate identity, and obtain written confirmation of the provider's controls (currently U). For **usability**, I restore monthly into an isolated sandbox, run database consistency checks, compare row counts and the latest transaction time, run application smoke tests, and obtain function-owner sign-off. Unverified assumptions at this stage are: **U1** B01 to B04 pass integrity checks (assumed only in the baseline drill); **U2** provider access independence; **U3** the hourly incremental pattern (A1); **U4** encryption keys are reachable during an outage; **U5** the contents and cleanliness of the 17:00 Day −3 backup.

Task 3: Recovery strategy and budget

I compared the packages against my RTO and RPO targets, treating the supplier timings as planning estimates that still need a witnessed test. Table 7 is the option matrix and Table 8 the annual cost.

Table 7: Option matrix

Criterion	Cold	Warm	Hot
First critical service	24 to 48 h	3 to 5 h (drill: LMS01 at 3 h 45 min)	About 1 h
Data-loss window	Up to 24 h	1 to 6 h	Up to 15 min

Against RTO and MTD	Fails F1, F2, F3 (MTD 8 to 10 h); borderline F4, F5	Meets F2 to F5; F1 within RTO only at the lower end of the range	Meets all on paper
Against RPO	Fails all (24 h versus 30 min to 8 h)	Supplier window cannot guarantee 30 min for F1, F2; my 15-minute backups can if restored there	Meets all on paper
Staffing	Rebuild from scratch; two technicians cannot do it quickly	Both technicians fully used; a cross-trained third person needed	Least manual work; failback and validation still need staff
Geography and suppliers	Few supplier dependencies, but slow to stand up	Must sit in a different flood and power zone; depends on supplier readiness	Same checks, plus replication link and a single supplier
Corruption or malicious change	Older backups lower the risk of copying bad data	Restores from backup, so can choose a clean point	Replica copies corruption fast; needs the independent backups
Annual cost (Table 8)	NGN 1,200,000	NGN 2,100,000	NGN 3,000,000

Table 8: Itemised annual budget (NGN)

Package	Base cost	Independent protected backups	Training and tests	Total	Headroom against 3,000,000
Cold	600,000	400,000	200,000	1,200,000	1,800,000
Warm	1,500,000	400,000	200,000	2,100,000	900,000
Hot	2,400,000	400,000	200,000	3,000,000	0

All three options fit the NGN 3,000,000 limit, though Hot fits only exactly, with nothing left for a price rise, an extra test or an incident. **I recommend Warm at NGN 2,100,000.** Cold cannot meet any of my F1 to F3 time targets. Hot meets every target on paper but consumes the whole budget and, as a live copy, would propagate corruption, so it still relies on the independent backups. Warm meets the RTOs for F2 to F5, and F1's RTO if the first service lands at the favourable end of its range, and its modelled recovery in Task 5 finishes everything by 12:15. The trade-offs are clear. Warm **cannot meet the 11:00 Day 0 examination** by restoration, so continuity procedures carry it. It also cannot promise the 30-minute RPO for F1 and F2 unless the supplier window tightens to one hour and my 15-minute backups are restored there. I would make that a contract condition.

Residual risks and acceptance: the 11:00 exam and the F1/F2 RPO gap are accepted by the Principal with the Head of Teaching and Assessment and the Finance Lead. The unvalidated supplier timings are accepted by the Principal with Procurement until a witnessed test is passed. The two-technician dependency is accepted by the Head of IT. Backup-compromise risk stays with the Principal as risk owner until the identity separation in Task 5 is complete.

Cost against disruption

Warm costs NGN 2,100,000 a year, against estimated disruption of NGN 1,200,000 at 8 hours and NGN 3,360,000 at 24 hours with no measures. For scale, the Task 5 schedule implies about NGN 505,000 of disruption (F1 3.75 h × 60,000 = 225,000; F2 3.5 h × 40,000 = 140,000; F3 4.25 h × 20,000 = 85,000; F4 4.25 h × 10,000 = 42,500; F5 2.5 h × 5,000 = 12,500), assuming the examination workaround is accepted. That avoids roughly NGN 695,000 against the 8-hour case, so an illustrative break-even would need about three such events a year. A single-outage estimate therefore cannot establish return on investment. ROI needs an annual frequency, a probability distribution of outage lengths, the share of loss each option actually avoids, and costs

the estimates omit, and the estimates themselves are exercise values. Much of the benefit is also non-financial: learner welfare, integrity and recovery from ransomware or flood events that could last far longer than 24 hours.

Before approving spending I want: a flood and outage risk assessment with local history; a witnessed failover test timing each action; contract terms with penalties for missed recovery times; written proof of the supplier's geographic and power independence and access controls; restore-test results for B01 to B04; function owners' validation of the BIA figures; and Finance's review of insurance and the budget.

Task 4: Business continuity and disaster recovery plan

Scope, activation and coordination

The plan covers loss or inaccessibility of the main site or any shared system that supports F1 to F5, for 90 staff and 1,800 learners. The **Principal** is the authorised decision maker and the **Director of Operations** is the deputy. If neither answers within 15 minutes, I can activate Levels 1 and 2 myself and inform them afterwards. Escalation then runs from me to the Principal, and from the Technical Lead to the Head of IT and the incident response (IR) lead. Contact roles are held in a sealed contact sheet at the alternate site and offline on two phones, because directory-based contact lists may not be available.

Table 9: Activation levels

Level	Threshold	Action
1 Alert	Threat to site or systems; a critical function at risk within 24 h	Coordinator briefs Principal; contacts verified; no activation yet.
2 Continuity activation	Main site or any F1 to F5 service unavailable and expected to stay so for more than 4 h, or a deadline within 8 h is at risk	Open incident cell; start temporary procedures; book alternate site.
3 Full recovery	Outage expected beyond 8 h, shared infrastructure lost, or suspected compromise	Start DR runbook; IR lead joins; Principal approves spending.

Communication uses Microsoft 365 email and Teams first, then SMS broadcast, then a phone tree, with personal hotspots as the backstop. Situation reports follow a fixed format (status, deadlines, issues, decisions, next update) hourly on Day 0, every four hours once stable, and daily after that. One incident cell, which I lead, joins four groups: the continuity team (function owners and communications), the technical recovery team (two technicians under the Technical Lead), Facilities, and IR. They share a single decision log. If compromise indicators appear, IR can pause restoration, and the Principal resolves any conflict between speed and safety.

Table 10: RACI (A = accountable, R = responsible, C = consulted, I = informed; A/R = combined)

Activity	Principal	BC Coord.	Tech Lead	Facilities	IR Lead	Function owners	Comms Lead
1 Plan activation	A	R	C	C	I	I	I
2 Temporary assessment delivery	I	C	R	R	I	A/R (Head of Assessment)	I
3 Backup selection and validation	I	I	A/R		C	I	
4 System restoration	I	C	A/R	C	C	C	I

5 Business acceptance of restored services	I	C	R		I	A/R	I
6 Stakeholder communication	C	R	C	C	C	C	A/R
7 Suspected-compromise response	I	C	R	I	A/R	I	C
8 Return to primary site	A	R	R	R	C	R	I

Temporary procedures for the 72-hour outage

The assessment problem is capacity. Of the 120 learners, 80 can use their own devices and 40 need a centre workstation, but the alternate facility offers only 25 stations from 10:00. My plan is that 80 sit the 11:00 exam through controlled Microsoft 365 delivery with remote supervision, and 25 of the 40 take stations at 11:00, with learners who have documented accessibility needs seated first. The remaining 15 sit a supervised later sitting at 13:00 on an equivalent version, arriving by 12:30 with phones surrendered, to protect integrity. The Head of Assessment must confirm that this counts as an acceptable service, because it decides whether the NGN 120,000 trigger applies. Table 11 covers the five functions.

Table 11: Temporary procedures during the outage

Function	Temporary procedure	Approvals and controls	Privacy and reconciliation
F1	Sittings as described; adjusted arrangements (extra time, assistive software, accessible room) honoured from the adjustment register (A2); remote candidates verified by ID check on Teams.	Head of Assessment approves each exam version; invigilator and attendance logs; submissions hashed and stored in a restricted folder; marks released only after LMS01 validation.	Share links only to registered learners; delete copies after import; compare submission list with register and LMS01.
F2	Provider portal confirms payments; fees entered in a restricted manual log; provisional confirmation sent by 16:00.	Preparer plus separate approver; no wallet changes or refunds until reconciliation.	Reconcile log, provider export and FIN01; investigate differences before final confirmation.
F3	Online form open to 18:00; registration changes logged and approved.	Registrar approves each change; no deletions; unique reference numbers.	Collect minimum data only; import into REG01 after validation and count entries match.
F4	Use restored HR01 once validated; fallback is the last approved payroll plus a signed change list.	HR manager and finance approver both sign; duplicate check against the last bank file.	Payroll files in restricted storage only; compare totals with restored HR01 after recovery.
F5	Request register in a restricted mailbox; urgent certificates only with approval.	Registrar approval; no files on personal devices; existing access limits unchanged.	Check issued items against restored FS01 and update records.

The minimum resources for this arrangement are:

- **Staff:** me as coordinator; two technicians; an assessment lead; two invigilators at the alternate site and one remote supervisor; a finance officer and a separate approver; one registry officer and the Registrar; an HR officer; and one records officer.
- **Equipment:** the 25 stations, ten spare locked-down laptops, chargers, a printer, printed exam versions, two phones and hotspots, and the sealed contact sheet.
- **Supplier support:** the alternate-site provider, the payment provider's reporting contact, Microsoft 365 support, the bank portal, and the offsite backup provider's retrieval line.

DR runbook

Table 12 sets out the runbook. Restoring technology means the systems run and pass technical checks. Resuming an acceptable business service is a later step, which happens only after the owner confirms the data is usable and the service meets the minimum level.

Table 12: DR runbook

Step	Action and prerequisite	Owner	Completion check	If the step fails
1	Confirm activation; open decision log (prereq: Level 3 decision).	BC Coord.	Cell opened; first SITREP sent.	Deputy decision maker activates.
2	IR triage: check for compromise indicators, including backup logons.	IR Lead	Written go/no-go for restoration on an isolated segment.	Treat as compromise; follow Step 2a (isolate, reset credentials, preserve logs).
3	Select backups: B01 plus B02 to 07:00, B03, B04 (prereq: Step 2).	Tech Lead	Catalogue checked against provider inventory.	Use the last intact chain link; as last resort the 17:00 Day −3 full (U5).
4	Recovery network (30 min).	Technician 1	Connectivity and DNS tests pass.	Switch to mobile hotspots; escalate to the provider.
5	Restore AUTH01 into an isolated segment (30 min).	Technician 1	Directory loads; admin accounts reviewed.	Rebuild from configuration export; reset privileged credentials.
6	Security and backup checks (30 min): checksums for every link, malware scan, logon-log review.	Technician 1 with IR	All checks pass.	Pause DB01; re-select earlier point; IR leads.
7	Restore DB01 (90 min): B01, then increments in order; run consistency check.	Technician 1	Consistency passes; last transaction about 07:00; row counts as expected.	Retry from earlier point; fall back to manual service.
8	Restore FS01 in parallel (60 min).	Technician 2	Sample files open; permissions match backup.	Delay F5; use request register.
9	Restore LMS01, FIN01, REG01, HR01 with smoke tests.	Technicians 1 and 2	Login and sample transactions succeed.	Roll back to the snapshot taken before the application restore.
10	Business acceptance: owners run their checks (fees versus provider export; registry sample; payroll totals; file sample).	Function owners	Signed acceptance per function.	Continue temporary procedure; Tech Lead fixes; retest.
11	Declare service resumed; record time against RTO.	BC Coord.	Principal informed.	Hold declaration.
12	Reconcile temporary records into restored systems.	Function owners	Counts and totals match.	Keep manual log as the record of truth until fixed.

Communication, incident response and return

Table 13: Stakeholder communication matrix

Stakeholder	Information needed	Channel	Owner	Timing
Learners	Closure, exam arrangements, deadlines, support	Email, SMS	Comms Lead	Notice by 08:45; updates 10:30 and every 4 h
Staff	Safety, work location, roles	Teams, email, SMS	BC Coord.	08:30, midday, end of day

Applicants	Admissions open online until 18:00	Website, email	Registrar	09:30
Payment provider	Availability; transaction exports	Phone, email	Finance Lead	09:00 and 12:00
Bank	Payroll timing	Phone, portal	HR and Payroll Manager	Day 0 afternoon
Alternate-site provider	Activation, access, stations	Phone	Facilities	08:15
Offsite backup provider	Catalogue, retrieval, access logs	Phone, secure portal	Tech Lead	08:15
Principal and Board	Hourly SITREP; decisions needed	Teams, phone	BC Coord.	Hourly Day 0
Insurer, data-protection regulator	Loss notice; breach if confirmed	Formal notice	Principal, IR Lead	As required

Initial learner notice (Day 0)

Important notice for all learners. Because of flooding and a power failure, our main campus is closed this morning, and we expect it to stay closed for about 72 hours. Your safety comes first. Today's 11:00 examination will still go ahead. If you booked a centre workstation, please go to our alternate facility by 10:30 with your student ID. If you are using your own device, check your institute email at 10:30 for sign-in instructions and make sure your device is charged. Fee confirmation and admissions will continue online. If you need an accessibility adjustment, reply to this message and we will arrange it. Please use only your institute email, not social media, for personal information. We will update you by email and SMS at least every four hours. We are sorry for the disruption and thank you for your patience.

Incident response joins immediately if I see any sign of possible compromise: an unexpected backup administrator logon, a failed integrity check that I cannot explain, unexplained changes or deleted backups, ransom notes, or use of the shared administrator identity outside a change window. IR then owns evidence handling and decides when restoration may continue, DR proceeds only on an isolated segment, and no restored system joins production until IR clears it (NIST, 2025).

Return to the primary site is safe only when five things are true. Facilities provide written clearance for structure, electrics, water damage and drying, with power stable for 24 hours. IT has inspected hardware, assumed nothing about equipment that got wet, and validated or rebuilt every system. Security confirms the restored environment is clean. New records created at the alternate site are transferred in a controlled way: freeze entries, export, hash, import into the primary site, then check counts and totals with two people, keeping the alternate-site copy read-only. Finally, function owners validate each service and the Principal gives final sign-off after reviewing Facilities clearance and business validation. I would return in phases.

Task 5: Recovery schedule and exercise

Modelled baseline schedule

Table 14 uses the current warm-site drill timings and the two-technician limit. It is a modelled baseline, not a live test. The assessment chain is strictly sequential: network (30), AUTH01 (30), checks (30), DB01 (90) and LMS01 (45) total 225 minutes, so **the earliest possible assessment service is 11:45**. Nothing can shorten this chain, because each action needs the one before. The second technician starts FS01 at 09:30 while DB01 runs.

After DB01, the remaining four actions need 150 minutes of work, so two technicians finish no sooner than 75 minutes later: **all five functions are feasibly complete by 12:15**.

Table 14: Baseline recovery schedule (Day 0)

Action	Technician	Start	Finish	Prerequisite
Recovery network	T1	08:00	08:30	None
AUTH01 restoration	T1	08:30	09:00	Recovery network
Security and backup checks	T1	09:00	09:30	AUTH01 restoration
DB01 restoration	T1	09:30	11:00	Security and backup checks
FS01 restoration	T2	09:30	10:30	AUTH01 and security and backup checks
LMS01 restoration	T1	11:00	11:45	DB01 restoration
FIN01 restoration	T2	11:00	11:30	DB01 restoration
HR01 restoration	T2	11:30	12:15	DB01 restoration
REG01 restoration	T1	11:45	12:15	DB01 restoration

Controlling dependency path

Recovery network → AUTH01 → security and backup checks → DB01 → LMS01, ending at 11:45.

Table 15: Modelled results against RTO, MTD and RPO

Function	Available	Elapsed	RTO (margin)	MTD (margin)	Restored data age versus RPO
F1	11:45	3 h 45 min	4 h: met (15 min)	8 h: met (4 h 15 min)	1 h versus 30 min: not met
F2	11:30	3 h 30 min	5 h: met (1 h 30 min)	8 h: met (4 h 30 min)	1 h versus 30 min: not met
F3	12:15	4 h 15 min	6 h: met (1 h 45 min)	10 h: met (5 h 45 min)	1 h versus 1 h: met, no margin
F4	12:15	4 h 15 min	8 h: met (3 h 45 min)	24 h: met (19 h 45 min)	1 h versus 4 h: met
F5	10:30	2 h 30 min	12 h: met (9 h 30 min)	36 h: met (33 h 30 min)	9 h versus 8 h: not met by 1 h

Successful restoration is not enough on its own, and four deadlines still need a workaround. The 11:00 exam comes 45 minutes before LMS01. Fee confirmation at 16:00 is met technically, but FIN01 lacks payments from before 08:00 onward, so confirmations stay provisional until reconciled. Admissions and registrations received between 08:00 and 12:15 must be captured manually and then entered. Payroll is safe on timing but still needs the approved change list reconciled before the 12:00 Day 1 submission. The F1 and F2 RPO misses and the F5 miss are also data issues that restoration cannot solve.

Tabletop exercise (45 minutes)

I would run this as a 45-minute tabletop with one facilitator (me) and simulated roles for the Principal, Technical Lead, IR Lead, Facilities Manager, Head of Assessment, Finance Lead and Communications Lead. Minutes 0 to 10 cover the briefing and activation, each inject then gets about 10 minutes, and minutes 40 to 45 end with a hot wash. This is a stress scenario kept separate from the baseline schedule, so the effects below are estimates for discussion, not new commitments.

Table 16: Exercise log

Min	Inject	Decision and revised action	Owner	Rationale	Effect

10	Closure extends from 72 hours to five days.	Move to extended operations: renew alternate-site booking to day 5; shift rota to shifts; take backups of the alternate environment on the Table 6 schedule; reschedule remaining exams that week.	BC Coord., Principal approves spend	Temporary procedures were designed for 72 h and fatigue and manual-error risk grew with duration.	Higher cost; recovery plan unchanged but must run for longer.
20	The alternate facility offers only 15 working workstations.	Seat 15 highest-need learners at 11:00 (accessibility first). The other 25 sit at 13:00 on the 15 stations plus 10 loaned laptops, or 13:00 and 14:30 if laptops are unavailable, on an equivalent version with supervised holding.	Head of Assessment	Keeps a same-day sitting for all while protecting integrity.	Exam service continues; NGN 120,000 trigger depends on the Head of Assessment accepting the arrangement.
30	A DB01 incremental fails its integrity check; an unexpected backup administrator logon needs investigation.	Stop using the chain beyond the last intact link. IR takes control: isolate backup server, reset the shared administrator credentials, preserve logs, keep restores on an isolated segment. Ask the offsite provider for independent logs. Fall back to an earlier point; last resort is the 17:00 Day −3 full (U5).	IR Lead, Tech Lead	A broken chain makes later data unreliable and the logon may signal compromise.	The data-loss window widens beyond 1 h. Illustrative effect: about one extra hour of checks puts DB01 at 12:00 and all applications at about 13:15, so F2 (12:30) still meets 16:00; FS01 waits for IR clearance.
40	Payment provider reports successful payments absent from restored finance records.	Treat the provider export as authoritative. Freeze receipt and wallet changes. Build a reconciliation ledger from 23:00 Day −1; each entry matched to FIN01 or the manual log, with two-person approval. Issue provisional fee confirmations by 16:00. Escalate any missing payment dated before the restore point as a possible integrity issue.	Finance Lead	Missing payments after 07:00 are expected from the RPO gap, but earlier ones could point to data loss or tampering linked to the inject at minute 30.	F2 meets deadline provisionally; final confirmation waits for reconciliation.

Table 17: Corrective actions

No.	**Corrective action**	**Priority**	**Owner**	**Target date**	**Completion evidence**
1	Separate backup administration from production identities; MFA; sealed break-glass; alert on backup administrator logon.	High	Head of IT	30 Nov 2026	Identity export shows separate accounts; alert test log; access review signed.
2	Restore-test the full DB01 chain, AUTH01 and FS01 in a sandbox with business validation.	High	Technical Lead	15 Dec 2026	Test report: 100% of chain links verified; row counts reconcile; owner sign-off.
3	Add an immutable or offline copy and obtain written proof of provider access independence.	High	Head of IT with Procurement	31 Dec 2026	Object-lock configuration record; signed provider statement; retrieval test with separate credentials.
4	Rehearse the assessment continuity arrangement, including an exam version for a later sitting.	High	Head of Teaching and Assessment	15 Jan 2027	Rehearsal with at least 80 remote candidates; integrity checklist closed with no major findings.
5	Move DB01 to 15-minute incrementals with 4-hourly differentials.	High	Database administrator	31 Jan 2027	30 days of job logs show recovery point age of 15 min or less.

6	Secure a contractual minimum of 40 workstations and confirm flood and power independence of the alternate site.	Medium	Facilities Manager	28 Feb 2027	Signed agreement with the capacity clause and a site risk note.
7	Cross-train a third person on the runbook.	Medium	Head of IT	31 Mar 2027	Training record and a drill in which the deputy completes a restore step.

Task 6: Governance and performance measures

The **BC/DR Coordinator owns the plan**. The Principal approves it, and the Head of IT, Facilities Manager, IR Lead and function owners review it each year and after every test or incident. Version control uses a numbered document with a change log, stored in a controlled repository, with printed copies in the alternate-site grab pack and offline copies held by role holders. New staff in BC/DR roles are trained within 30 days, and everyone in a role has an annual refresher, with extra technical training for technicians.

Table 18: Risk-based testing calendar

Activity	Frequency	Risk basis or trigger	Owner
Sandbox restore of DB01; backup success report	Monthly	Highest-risk shared asset	Technical Lead
Full-chain, AUTH01 and FS01 restore test; call-tree test	Quarterly	Untested chains are unproven	Head of IT
Tabletop exercise (one before the rainy season)	Twice a year	Flood season and role changes	BC Coord.
Warm-site failover exercise with function owners	Annually	Validate supplier timings	BC Coord.
Unscheduled review and retest	Event-driven	Major system, supplier or site change; incident; audit finding; flood warning	BC Coord.

The mapping below is an educational alignment, and I do not claim certification. In Table 19 I map my outputs to the seven NIST SP 800-34 planning steps and, in Table 20, to the ISO 22301 clause themes (NIST, 2010; ISO, 2019).

Table 19: NIST SP 800-34 Rev. 1 planning steps

NIST step	Output in this report
1 Contingency planning policy	Scope, activation levels, decision maker and ownership (Task 4, Task 6)
2 Business impact analysis	BIA and impact calculations (Task 1); objective matrix (Task 2)
3 Preventive controls	Backup design, identity separation, flood measures (Task 2, Task 6)
4 Recovery strategies	Option comparison and Warm recommendation (Task 3)
5 Contingency plan	RACI, temporary procedures, DR runbook, communications (Task 4)
6 Testing, training and exercises	Baseline schedule, tabletop, calendar (Task 5, Task 6)
7 Plan maintenance	Review triggers, version control, corrective actions (Task 5, Task 6)

Table 20: ISO 22301:2019 themes

ISO theme	Output in this report
Context of the organisation	Scenario, assumptions, flood consideration
Leadership	Principal as decision maker; ownership and approval

Planning	Objectives, risk acceptance, corrective actions
Support	Resources, training, communication matrix, document control
Operation	BIA, strategies, runbook, temporary procedures, exercise
Performance evaluation	KPIs and testing results
Improvement	Corrective actions and the 12-month maturity target

Flood and climate consideration. The 2024 amendment to ISO 22301 asks organisations to decide whether climate change is a relevant issue for their continuity system (ISO, 2024). For Northbridge, flooding is a physical risk that has already struck. I would check the alternate site and the offsite repository for flood exposure and shared power so that they cannot fail together with the main site, move critical equipment above flood level, and add a rule that official flood warnings trigger Level 1 and pre-emptive backups.

Table 21: Key performance indicators

KPI	**Formula or method**	**Data source**	**Target**	**Frequency**	**Owner**
Recovery performance	Functions recovered within RTO ÷ functions tested × 100; RPO achieved shown separately	Exercise log, runbook timings	100% for F1 to F4; RPO met in 4 of 5 functions by Mar 2027	After each test, quarterly report	Technical Lead
Restoration-test coverage	Backup sets (B01 to B04, plus new sets) with a successful restore test in last 90 days ÷ total × 100	Restore-test reports	100%	Monthly	Head of IT
Corrective-action closure	Actions closed by target date ÷ actions due × 100	Action register	90% on time; 100% of High actions	Monthly	BC Coord.
Training	Role holders trained in last 12 months ÷ total role holders × 100	Training records	95%	Quarterly	HR and Payroll Manager

Maturity. I use a five-level scale (1 Initial, 2 Developing, 3 Defined, 4 Managed, 5 Optimised) and read it against the slide 13 model (ICDFA, Week 21). Northbridge is at **Level 2, Developing**: there are plans and backups, but recovery targets are untested, backup integrity and access independence are unverified, and one identity covers production and backup administration. A realistic 12-month target is **Level 3, Defined**: objectives tested and approved, restore tests running on schedule, corrective actions closed and roles trained. I would aim for Level 4 only for backup verification, where KPI data can be measured monthly, and not claim it for the whole programme.

Declaration and AI acknowledgement

I used an AI assistant (Claude, by Anthropic) to help structure and draft this report, in line with the course AI policy. I reviewed the calculations and decisions myself and I am ready to explain and defend them. The scenario data are fictional, and the framework mapping is educational and does not claim certification.

References

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